SYSTEM ANALYSIS DOCUMENTATION (SAD)

UMHackathon 2026

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1. *Introduction:*

1.1. *Purpose:*

The HalalChain Agent is designed to solve the "Compliance Bottleneck" faced by SMEs in Malaysia when applying for JAKIM Halal certification. The system automates the ingestion, classification, and verification of complex document bundles that are traditionally processed manually.

1. ***Architecture**:*

   The system utilizes a **Client-Server Architecture** designed for scalability:

   - **Frontend:** React/Tailwind CSS dashboard for SMEs and a simplified portal for Suppliers.
   - **Backend API:** Next.js serving as the orchestration layer between the UI and the AI.
   - **State Management:** localStorage (Demo) to maintain the workflow state (Upload → Verify → Resolve → Export).
   - **AI Engine: Gemini AI** acts as the "Central Brain" for all reasoning tasks.

2. ***Data Flows:***
   a. **Ingestion:** User uploads unstructured files.
   b. **Analysis:** API Route sends file metadata and text for classification.
   c. **Verification:** AI parses ingredients and cross-references them against JAKIM standards.
   d. **Resolution:** If an ingredient is "Ambiguous" (<70% confidence), the system triggers a Supplier Loop.

3. ***Model Process:***
   a. **Ingestion & Interpretation Layer:** The process begins when the user uploads a batch of unsorted documents. The system performs a "Read-Understand-Classify" loop where the Gemini AI identifies the document type (e.g., Premises License vs. Ingredient List) and auto-corrects the file naming convention.
   b. **Compliance Audit Layer:** The model cross-references the identified documents against a mandatory JAKIM checklist. If gaps are detected, the system branches into a "Missing Document" state, alerting the user.
   c. **Reasoning & Verification Layer:** For ingredients, the model executes a multi-step verification process:
      i. **Normalization:** Mapping local/trade names to scientific terms.
      ii. **Risk Assessment:** Gemini AI determines Halal status based on JAKIM guidelines.
      iii. **Ambiguity Resolution:** If confidence is <70%, the model triggers an external branch (the Supplier Orchestration Loop) to request missing evidence.

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d. **Synthesis & Export Layer:** Once the "All-Clear" state is reached, the model aggregates the audit trail, supplier responses, and verified lists bundle ready for the MYeHALAL portal.

4. ***Role of Reference:***
   a. **For Developers (frontend/backend):** it defines the state machine transitions (e.g., moving from Ambiguous to Verified) and the API schema required for communication between the FastAPI backend and the React frontend.
   b. **For Prompt Engineers:** It outlines the specific reasoning logic the Gemini AI must follow, ensuring that the "Reasoning Audit Trail" matches the technical expectations for Malaysian regulatory compliance.
   c. **For Testers/Judges:** It provides a high-level architecture map to verify that the system is truly "AI-Native"—demonstrating that if the Gemini AI reasoning engine were removed, the system would lose its ability to handle unstructured data or navigate regulatory ambiguity.
   d. **For Scalability Planning:** It acts as a blueprint for future integrations, such as replacing the browser-based text extraction with a full-scale OCR microservice or integrating with a live JAKIM database.

*1.2. **Background**:*

SMEs often struggle with unstructured data (photos of licenses, handwritten ingredient lists) and ambiguous regulatory requirements. HalalChain uses an AI-first approach to transform this "messy" input into a structured, JAKIM-ready digital twin.

1.* Previous Version*:

The current landscape of Halal certification management for SMEs is characterized by fragmented, manual processes:

- **Manual Document Filing:** Existing solutions are primarily cloud-storage repositories (like Google Drive or physical folders) where users must manually name, categorize, and sort documents according to JAKIM's requirements.
- **Static Checklists:** Current "digital assistants" are simple static forms. They cannot "read" the content of an uploaded file to verify if a "Premises License" is actually a license or just a blank receipt.

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- **Human-Dependent Verification:** Ingredient checking requires a Halal Executive to manually cross-reference every single item against the JAKIM Sistem Maklumat Halal or supplier certificates, leading to high human error and long lead times.
- **Disconnected Supplier Communication:** When information is missing, SMEs rely on unsynchronized email threads or WhatsApp messages to chase suppliers for certificates, which are often lost or never integrated back into the main application.

2. ***Changes in Major Architectural Components***

The **HalalChain Agent** introduces an "AI-Orchestration Layer" that shifts the system from a passive storage tool to an active regulatory agent.

AI-Driven "Content-Aware" Ingestion

- **Old:** User uploads file123.pdf -> User manually selects "Premises License" from a dropdown.
- **New:** User drops file123.pdf -> **Gemini AI** reads the text, identifies the document type, renames it to Premises_License_2026.pdf, and checks the expiry date automatically.

Multi-Step Reasoning Engine (The "Brain")

- **Old:** Hard-coded keyword matching for ingredients (e.g., if "Gelatin" exists, flag as "Check").
- **New:** The AI performs **Semantic Reasoning**. It understands that "Minyak Sapi" is a local term for beef tallow and evaluates its Halal status based on the source and processing method described in the uploaded text, rather than just the name.

Autonomous State-Machine for Gaps

- **New Capability:** The system maintains a dynamic "Workflow State." If a critical document is missing or an ingredient is marked as Ambiguous, the architecture automatically triggers a **Supplier Resolution Loop**. It generates the API request and the external portal link without developer intervention.

Structured Regulatory Synthesis

- **Old:** User prints out a list of files to hand-carry or upload one by one to MYeHALAL.
- **New:** The system aggregates the entire decision trail—including the Gemini AI reasoning for every ingredient—and synthesizes it into a single **JSON Schema** specifically mapped to JAKIM's digital requirements.

3. ***New capabilities that are only introduced in this***

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- **Automated Naming Convention Enforcement:** High-accuracy file tagging using LLM vision/text analysis.
- **Confidence-Based Logic:** A risk-mitigation layer that only allows "High Confidence" items to pass, while forcing human/supplier intervention for "Low Confidence" items (<70%).
- **Reasoning Audit Trail:** A technical log that explains *why* the AI made a certain decision (e.g., "Ingredient X is flagged because it contains potential bovine derivatives from a non-certified region").
- **One-Click JAKIM Export:** The ability to bundle complex, multi-source data into a standardized digital format compatible with government portals.

1.3. Target Stakeholder

Stakeholders	***Roles***	***Expectations***
SME Business Owner (Applicant)	Uploads raw documents (invoices, licenses, ingredient lists) and manages the Halal application workflow.	A simplified, “zero-mess” experience where AI handles document naming and identifies missing requirements automatically
Suppliers	Receives automated verification requests for ambiguous ingredients and provides supporting Halal evidence.	A straightforward portal to declare ingredient status and upload certificates without needing a full account or complex training.
Regulatory Body (JAKIM/Auditors)	Receives the structured JSON/PDF export bundle generated by the system.	High-integrity data that follows official schemas, complete with a clear audit trail of how each ingredient was verified.
Development Team	Builds and maintains the FastAPI orchestration layer and Gemini AI prompt templates.	Modular code, clear state-machine logic for the "Ambiguity Loop," and well-defined Gemini AI contracts
QA Team	Validates the accuracy of the AI classification and the reliability of the supplier feedback sync.	Defined test cases for "edge-case" ingredients (e.g., Manglish terms), schema validation for the JAKIM export, and API failure handling.

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2. System Architecture & Design

2.1. High Level Architecture

2.1.1. Overview:

Type	Details
System	Web-based Dashboard (SME & Supplier Portals)
Architecture	Cloud-Native Microservices with AI-Orchestration Layer

HalalChain is a client-server system where two primary clients—the **SME Dashboard** and the **Supplier Portal**—interact via a **FastAPI** backend. The system is decomposed into three core microservices: **Document Service** (classification), **Verification Service** (ingredient reasoning), and **Orchestration Service** (supplier loops).

2.1.2 LLM as Service Layer

The architecture of your product needs to show how Gemini AI is integrated as a service layer- not limited to a generic "AI Module". The following Dependency diagram includes:

2.1.3 Dependency Diagram

- ***Prompt Construction:*** *Prompts are dynamically generated using* ***Pydantic templates****. Context includes the extracted file text, current compliance status, and Malaysian-specific regulatory guidelines.*
- ***Context Window Management:*** *To avoid token overflow, inputs are* ***chunked*** *(e.g., long ingredient lists are split into batches of 10). The system uses a* ***Recursive Character Text Splitter*** *before sending data to the Gemini AI.*
- ***Parsing & Flow:*** *The system enforces* ***Structured Output (JSON)****. The backend parses the response, validates the schema, and updates the PostgreSQL state. If confidence is <70%, the Orchestration Service automatically triggers a call to the Supplier Portal.*
- ***Major API Calls:***
- ***Frontend*** *->* ***AWS API Gateway*** *->* ***FastAPI****: Initial file upload and state fetch.*
- ***Verification Service*** *->* ***Gemini AI****: Reasoning for ingredient safety.*
- ***Orchestration Service*** *->* ***Amazon SES****: Emailing suppliers for missing certificates.*

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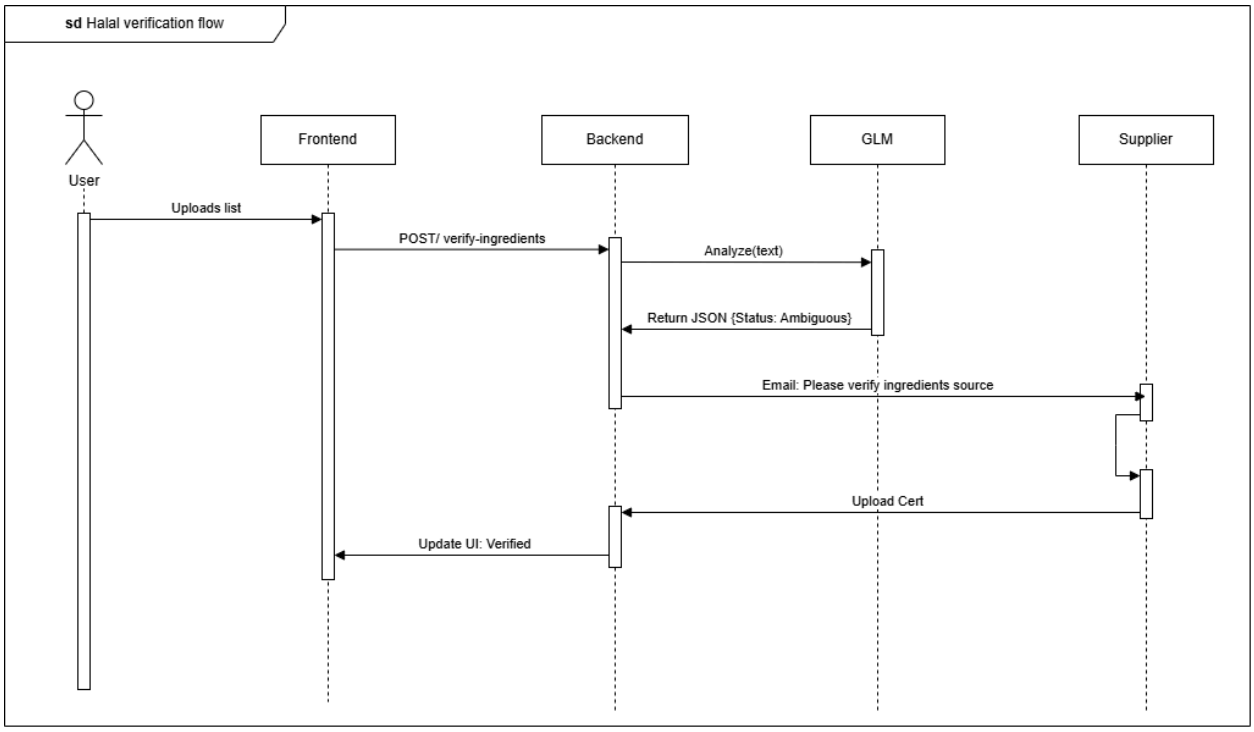
2.2.2. Sequence Diagram

This walk-through demonstrates the end-to-end journey of an ingredient verification:

1. **SME User:** Uploads an unstructured PDF "Ingredient List."
2. **Frontend:** Extracts text via FileReader and POSTs to /analyze-ingredients.
3. **Backend:** Sends text to **Gemini** to identify and normalize items (e.g., "Minyak Sapi" -> "Beef Tallow").
4. **Gemini AI:** Returns a JSON status (Halal/Ambiguous) and a reasoning string.
5. **System:** Sends an automated link to the **Supplier**.
6. **Supplier:** Uploads a certificate -> Backend updates state -> SME Dashboard reflects "Verified" status.



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2.2. Technological Stack

2.2.1. **Frontend: React.js** with Tailwind CSS (Responsive web dashboard for complex data management).

2.2.2. **Backend: FastAPI**

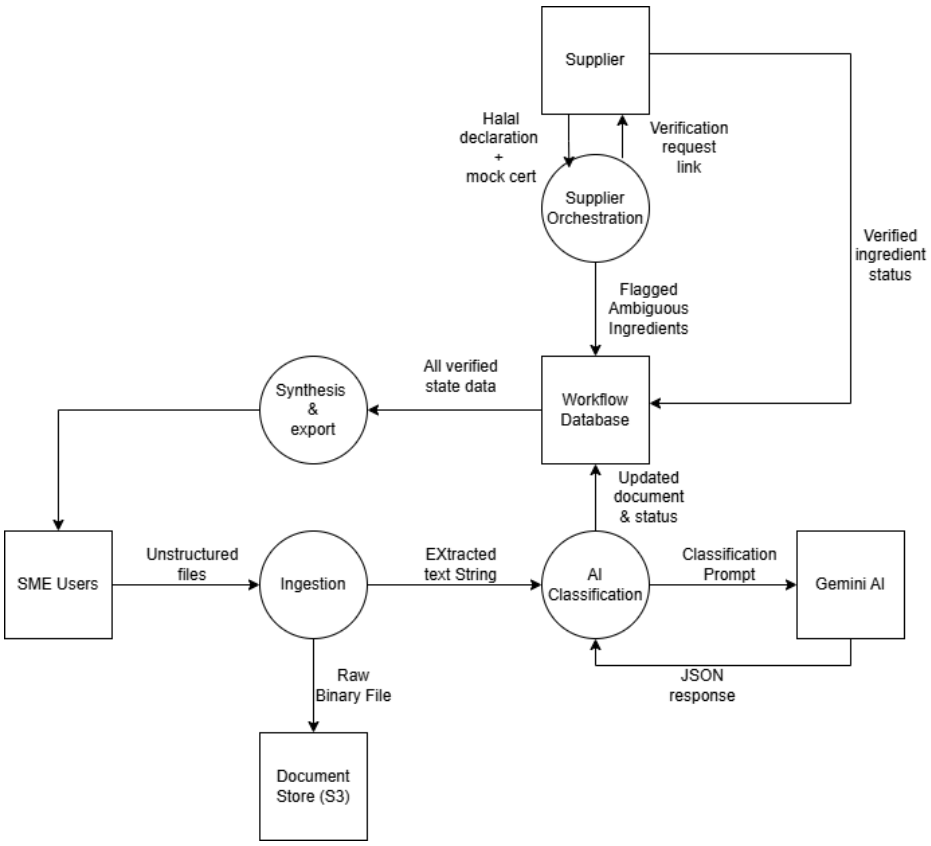
2.2.3. **Cloud/Deployment: AWS EC2** (App Server), **S3** (Document Storage), **Gemini AI** (AI Service Layer).

2.2.4. **Database: PostgreSQL (AWS RDS)** (Relational storage for complex mappings between SMEs, Ingredients, and Certificates).

2.3. Key Data Flows

2.3.1. Here, do show the how data are moving through the system and how its structure and stored

2.3.1.1. Data Flow Diagram (DFD)



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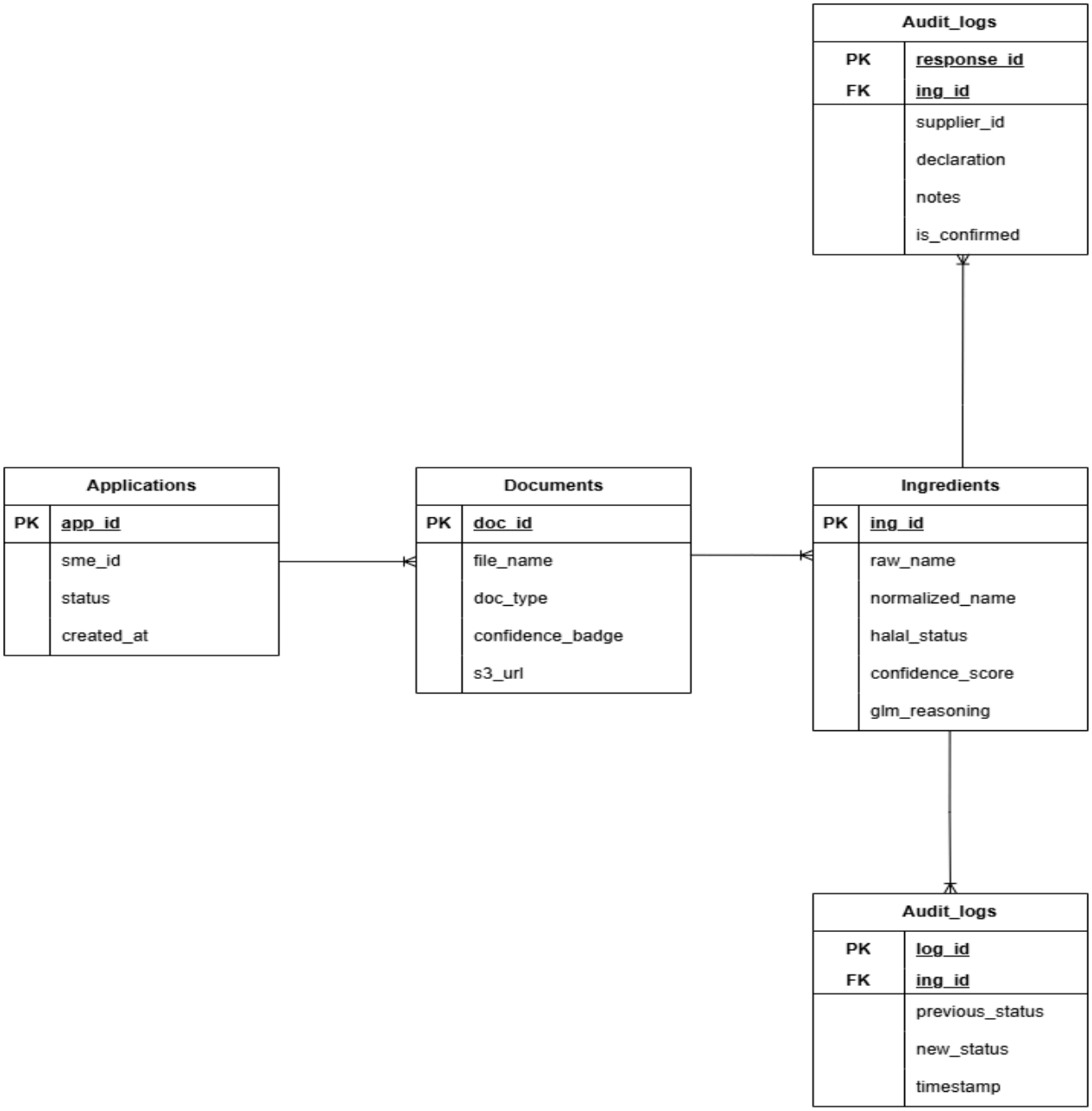
2.3.2. Normalized Database Schema



The Many-to-Many Relationship: one **Ingredient** might require verification from multiple **Suppliers**, and one **Supplier** might provide declarations for many different **Ingredients**

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3. Functional Requirements & Scope

This section highlights the core boundaries of the projects. That enables them to focus on a high-impact MVP to showcase the technical feasibility of the project.

3.1. Minimum Viable Product:

3.1.1. *Do list down 3-5 core features that the team is going to build and do a demo by the end of the UMHackathon 2026.* ***[e.g. in Table]***

#	Features	Description
1	**Smart Document Auto-Sort**	Users drag-and-drop unstructured files (PDF, JPG, Excel). The system uses Gemini AI to extract text and automatically classify documents into types like "Premises License" or "Ingredient List" with a confidence badge.
2	**Compliance Gap Checker**	The system cross-references uploaded files against the 4 mandatory JAKIM document requirements. It displays a real-time checklist and a warning banner if critical evidence is missing, blocking final export until resolved.
3	**AI Ingredient Verification**	Automatically parses ingredient names from uploads and uses AI reasoning to classify them as Halal, Haram, or Ambiguous. It provides a detailed "Reasoning Audit Trail" explaining the logic behind each status
4	**Supplier Resolution Loop**	For ingredients flagged as "Ambiguous" with low confidence (<70%), the system triggers an automated workflow. It generates a pre-filled verification request for suppliers to provide a manual declaration or upload a Halal certificate

3.2. Non-Functional Requirements (NFRs)

Here, highlights the system qualities that are not features but are crucial for reliability and improve the system. [e.g. table in below]

Quality	Requirements	Implementation

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Scalability	Prompt Chunking	Large document sets are divided into smaller "analysis chunks" to stay within AI token limits.
Reliability	State Persistence	All verification steps are saved to prevent data loss on page refresh.
Security	Data Privacy	Supplier portal is isolated; SMEs only see the "Verified/Declined" status of the ingredient.
Performance	Token Latency	AI prompts are optimized for speed; a 3-5 second response window is expected.

3.3. Out of Scope / Future Enhancements

Do identify the high-level features that the team planned for "Grand Vision" but not part of this UMHackathon 2026. [e.g. below]

a. **Integrated Communication & Collaboration**: Implementation of an in-app chat system to allow real-time clarification between SMEs and Suppliers regarding ingredient documentation.

b. **Marketing & Financial Incentives**: Development of a module for promotional offers, discount codes, and loyalty vouchers to encourage SME adoption of the platform.

c. **Advanced Scheduling & Renewals**: A "Scheduled Submission" feature that allows SMEs to prepare applications in advance and set automated reminders for Halal certificate renewals.

d. **Live Regulatory Integration**: Transitioning from a manual JSON/PDF export to a direct API integration with the JAKIM MYeHALAL portal for one-click submission.

e. **Vision-AI Certificate OCR**: Moving beyond text-based extraction to advanced AI capabilities that can scan physical, stamped Halal certificates and verify authenticity via QR code scanning.

f. **Global Halal Database Sync**: Real-time cross-referencing with international Halal certification bodies (e.g., MUI Indonesia, MUIS Singapore) to automate verification for imported ingredients.

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4. Monitor, Evaluation, Assumptions & Dependencies

4.1. Technical Evaluation:

4.1.1. ***Grayscale Rollout & A/B Testing***: New features, such as the "Manglish/Local Term Normalization," will be released to 10% of SME users initially to monitor AI reasoning accuracy. Once the confidence threshold of 70% is consistently met without manual overrides, the rollout will gradually increase to 100%. A/B testing will be used to compare different prompt engineering strategies for ingredient verification to determine which yields higher accuracy for complex Malaysian trade names.

4.1.2. ***Emergency Rollback (ER) & Golden Release***: If the system detects a failure rate exceeding 2% in document classification or if the API latency exceeds 5 seconds (P95), an Emergency Rollback is triggered. The automated CI/CD pipeline will immediately deploy the "Golden Release"—a stable version of the application that allows for manual file tagging while the AI service is being restored.

4.1.3. Priority Matrix:

4.1.3.1.

Priority	Condition	Required Action
P1 - Critical	Gemini AI total outage or >5% verification error rate.	Trigger Emergency Rollback to Golden Release; enable manual override mode for all users.
P2 - High	Supplier Loop latency > 10 seconds.	Restart asynchronous notification workers and check email relay health.
P3 - Medium	Incorrect confidence badge assigned to >10% of files.	Refine prompt templates and re-run batch classification tests.

4.2. Monitoring:

4.2.1. **AI Success Rate:** Percentage of ingredients successfully classified without manual intervention.

4.2.2. **API Latency:** Monitoring the 3s response target for Gemini AI calls.

4.2.3. **State Integrity:** Ensuring localStorage sync matches to prevent data loss across user refreshes

4.3. Assumptions: *State key operational & environmental conditions that you assumed to be valid while development of system and deployment.*

a. Internet Stability: Users (SMEs and Suppliers) have stable internet connections to upload multi-megabyte PDF/JPG documents.
b. Document Legibility: Uploaded images and scans are of sufficient quality for text extraction (FileReader) to accurately parse ingredient lists.
c. Regulatory Context: It is assumed that the Gemini AI possesses the necessary training data to understand basic Malaysian Halal standards and common local food terms

4.4. ***External Dependencies:*** *Do list all the external tools that are applied to your system to function. [e.g. below]*

Tools	*Purpose*	*Risks*
Gemini AI	Core LLM reasoning engine for multi-step ingredient verification and structured JSON export.	**High Risk:** Rate limiting or downtime. **Mitigation**: Implement graceful degradation allowing manual user declaration
Browser LocalStorage	Serves as the primary mechanism for **State Persistence** in the prototype, maintaining the workflow state (Upload → Verify → Resolve → Export) without a live database	**Low Risk:** Data is cleared if the user manually wipes browser cache or switches devices.

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5. *Project Management & Team Contributions*

The duration of the Preliminary round is approximately 10 days considering the two-week sprint.

5.1. *Project Timeline:*

- **Day 1-3**: Project Planning & brainstorming.
- **Day 3-5**: Backend & AI Integration: Engineering the core prompts for document classification and ingredient verification.
- **Day 6-8**: Frontend & Supplier Loop: Developing the drag-and-drop UI and the Supplier Resolution portal for ambiguous items.
- **Day 9:** Testing & Optimization
- **Day 10:** Final Submission: Compiling the system documentation and recording the end-to-end workflow demo.

5.2. *Team Members Role:*

Syuhada
- user mapping
- user interface
- Quality Assurance Testing Documentation

Maiza
- supplier mapping
- supplier interface
- Product Requirement Document

Mujahid
- mock files (5)
- mock jakim api
- System Analysis Documentation

Rauf
- backend orchestration

5.3. *Recommendations:*

Scalability: Utilize **Redis Cache** for frequently checked ingredients (e.g., standard additives) to reduce redundant API costs and lower latency.

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Reliability: Implement **asynchronous task queues** (e.g., Celery) for the Supplier Resolution Loop to ensure that thousands of supplier emails do not block the main application thread.

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